

Ammar Yasir Naich

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SUMMARY

Applied AI Researcher and Engineer with a PhD in Artificial Intelligence with 5+ years, specialising in real-time computer vision, GPU/Edge AI, and agentic AI systems. Experienced in building and deploying production AI platforms using NVIDIA technologies, deep learning, LLMs, and scalable ML infrastructure.

SKILLS

- **Applied AI & ML:** Deep Learning, Transformers, Computer Vision, Multimodal AI, LLMs, Fine-tuning
- **Edge AI & Computer Vision:** NVIDIA DeepStream, Jetson Orin, CUDA, Multi-camera Analytics, Object Detection, Tracking
- **Agentic AI & RAG:** RAG, Agentic Workflows, Tool Orchestration, Memory, Grounding, Guardrails
- **AI Engineering & Deployment:** PyTorch, Hugging Face, Kubernetes, Kubeflow, MLflow, AWS, CI/CD
- **Programming & Optimisation:** Python, C++, SQL, GPU Acceleration, Low-latency Inference, Quantization

AWARD/CERTIFICATES

- **Best Industrial Paper Award:** International Conference on Deep Learning Theory and Application (DeLTA) 2026.
- **Nebius Fellowship Award:** Selected for the AI Inference Engineer track, Awarded the LLM Architecture credential for demonstrating expertise in large language model inference, deployment, optimization, and scalable serving infrastructure.
- **NVIDIA DLI:** Fundamentals of Accelerated Computing with CUDA Python.
- **PhD Research Scholarship:** Received the National Award for PhD Research from the Higher Education Commission of Pakistan.
- **Coursera, University of California San Diego:** Big Data, Version 1 Specialization.
- **Coursera, University of Toronto:** Visual Perception for Self-Driving Cars.

PROFESSIONAL SERVICE

- **Peer Reviewer:** IEEE Open Journal of the Computer Society
- **Conference Reviewer:** ICVES (International Conference on Vehicular Electronics and Safety)

PUBLICATIONS

- **Published:** LiDAR-based intensity-aware outdoor 3D object detection. Sensors 24.9 (2024): 2942.
- **Published:** PointConViT: Efficient 3D Object Detection with Adaptable Transformer Convolution Fusion, **Best Industrial Paper Award**, DeLTA (2026)

EXPERIENCE

- **Cold Banana** Bournemouth
AI Engineer (KTP Associate) *August 2026 - Present*
 - **Real-Time Edge AI:** Leading the development of a GPU-accelerated multi-camera analytics platform using NVIDIA DeepStream and Jetson Orin, integrating detection, tracking, facial expression, and gesture recognition for real-time edge deployment.
 - **Agentic AI & RAG:** Architecting an enterprise agentic RAG platform combining TurboVec, LLMs, tool orchestration, memory, grounding, and guardrails to build reliable context-aware AI workflows.
 - **End-to-End AI Engineering:** Driving AI systems from architecture and rapid prototyping through performance optimisation, benchmarking, integration, and production deployment.

- **Digital Reality Corp (DRC)** London
Machine Learning Engineer *September 2025 - October 2025*
 - **Production AI and Data Pipelines:** Developed and trained models converting LiDAR point clouds into 2D/3D digital assets, while optimising models and automating data pipelines with CI/CD and AWS services for scalable, reliable delivery.
- **Queen Mary University of London** London
Research Fellow *October 2021 - Present*
 - **International Teaching Nomination:** Nominated by QMUL to lecture at QUPT in Hainan, China, for the module *Introduction to Data Science*.
 - **Research Software and AI Evaluation Platform:** Leading development of a scalable AI platform for ML model hosting and evaluation on OpenStack, supporting organised model execution, reproducible evaluation workflows, and distributed training/deployment across GPU resources. Built internal tooling, CLI-based interfaces, and authentication-integrated research software for reliable experimentation and collaboration.
 - **Big Data Processing and Distributed Systems Teaching:** Developed coursework and laboratory exercises for Big Data Processing, including Apache Spark Streaming and Apache GraphFrames, while managing lab delivery, resolving technical queries, and assessing student work.
 - **Research Supervision and Communication:** Supervised 23 MSc students in deep learning and computer vision projects, supporting research communication, technical writing, and presentation of results.
 - **Machine Learning Teaching:** Designed and delivered advanced coursework and lab sessions in Principles of Machine Learning, integrating crowdsourced datasets and improving student competency in machine learning techniques by 75%.
- **Queen Mary University of London** London
PhD Research *June 2019 - July 2024*
 - **Efficient AI for Large-Scale Sensor Data:** Developed and evaluated deep learning models for real-time 3D object detection using large-scale datasets including KITTI, nuScenes, and Waymo, building reproducible training and evaluation pipelines for data-intensive experimentation.
 - **Optimisation for Resource-Constrained Systems:** Implemented custom CUDA kernels and leveraged model, data, and pipeline parallelism to improve computational efficiency across local and cluster GPU environments.
 - **Model Quantization and Edge-Oriented LLM Work:** Explored quantization strategies for efficient training and fine-tuning of LLM models under edge and low-memory GPU conditions, aligning with efficient inference and deployment objectives.
- **NodeNS** London
Embedded Software Engineer *January 2020 - January 2021*
 - **Edge Device Integration:** Built a sensor integration unit enabling plug-and-play connections between mmWave radar sensors and edge devices, supporting reliable data exchange in embedded environments.
 - **Secure Systems Engineering:** Designed and implemented a security protocol to support secure communication across the networked system.
 - **Engineering Tooling:** Defined the development technology stack and built a GUI tool for sensor data transfer and configuration, improving team productivity and deployment readiness.
- **Stingray Technologies (Pvt) Ltd (formerly Maritime System Limited)** Pakistan
Technical Manager / Software Architecture *January 2015 - September 2018*
 - **Software Architecture and Team Leadership:** Led software architecture and cross-functional engineering teams to deliver scalable, low-latency systems in C/C++/Qt across critical applications.
 - **GIS Application Development:** Led the development of a high-performance GIS application in C/C++/QGIS, integrating real-time tracking, data recovery, pattern matching, and data mining to enable advanced visual geospatial analysis and actionable insights, significantly enhancing service offerings and client engagement, and becoming the company's main source of revenue.
 - **Sensor and Diagnostic Data Systems:** Developed software for acquiring and processing real-time data from digital I/O and serial sensors, improving data processing speed by 40% and supporting scalable system integration.
 - **Simulation and System Validation:** Architected emulators and simulators in C/C++ for hardware system testing, improving software reliability and reducing project cost by 35%.
 - **Fault and Event Management:** Contributed to Critical Event Management Services for timely notification and response to 250 possible system events, reducing downtime by 70% and improving operational safety.

EDUCATION

- **Queen Mary University of London** London, UK
PhD in Computer Science *June 2019 - July 2024*

- **Mehran University of Engineering and Technology**
Master of Engineering in Networking
- **Mehran University of Engineering and Technology**
Bachelor of Engineering in Computer Systems

Jamshoro, Pakistan

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